U-Pb and Sm-Nd ages for the Stillwater Complex and its associated sills and dikes, Beartooth Mountains, Montana: Identification of a parent magma?

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ABSTRACT

Seven analyses from two samples of the Lower Banded series of the Stillwater Complex, Montana, yielded a U-Pb zircon-baddeleyite age of 2705 ±4 Ma. A more precise age of 2704 ±1 Ma is obtained if we regress only five of seven analyses. In either case, the age is in very good agreement with the previously reported Sm-Nd mineral isochron age of 2701 ±8 Ma for a gabbro from the Lower Banded series.

Noncumulate mafic sills and dikes that immediately underlie the Stillwater Complex are separated into five compositional groups. U-Pb zircon ages for a group 1 gabbronorite (2711 ±1 Ma) and a group 3 mafic norite (2703 ±10 Ma) are comparable to previously reported U-Pb zircon ages for a group 1 diabase (2713 ±3 Ma) and a group 4 high-Ti norite (2712 ±3 Ma). Although group 2 high-Mg gabbronorite and group 6 olivine gabbro have not yet yielded significant amounts of zircon, baddeleyite, or sphene, mineral separates from group 2 yielded a four-point Pb-Pb internal isochron age of 2704 ±25 Ma, and separates from group 6 yielded a Sm-Nd internal isochron age of 2706 ±64 Ma, indicating that these groups are at least contemporaneous with intrusion of the Stillwater Complex.

These results indicate that groups 1 and 4 of the associated sills and dikes are perhaps slightly older than the complex and that groups 2, 3, and 6 may be coeval with it, supporting the idea that these three groups are the best candidates for a Stillwater parent magma, a conclusion previously presented on the basis of detailed mineralogic and geochemical evidence.

INTRODUCTION

The Stillwater Complex, exposed along the northern flank of the Beartooth Mountains of southwestern Montana, is an Archean stratiform mafic to ultramafic intrusion that has been the focus of many isotopic studies concerned with crystallization processes, magma genesis, and mantle evolution (Czamanske and Zientek, 1985). The complex is divided into five major series (Fig. 1): the Basal series, predominantly orthopyroxene cumulates; the Ultramafic series, containing various proportions of cumulus olivine, bronzite, and chromite; and the Lower, Middle, and Upper Banded series, all of which contain cumulus plagioclase (Zientek et al., 1985). Rocks of the basal series are known to intrude the metamorphic rocks that underlie the complex (Page and Zientek, 1985).

A major problem confronting investigators of the complex is the lack of an unequivocal age of emplacement, a problem attributable to either source heterogeneities, contamination, postemplacement disturbances of the isotopic systems, or some combination of these factors. Prior to this study, a Sm-Nd mineral isochron age of 2701 ±8 Ma from a gabbro of the Lower Banded series (DePaolo and Wasserburg, 1979) was generally accepted as the age of the Stillwater Complex. This age, however, is in conflict with a Pb-Pb isochron age of 2662 ±12 Ma defined by data from seven whole-rock samples of various rock types of the cumulate stratigraphy (Manhes et al., 1980). Numerous other ages obtained by using a variety of radiometric techniques (Lambert et al., 1985, 1989) are not precise enough to resolve this discrepancy.

Diabasic rocks near the base of the Stillwater Complex, generally assumed by early investigators (Peoples, 1932; Hess, 1960; Jackson, 1971) to be a chill zone, are now recognized as one of several compositionally distinct magma types found as sills and dikes. These noncumulate intrusions, termed the "Stillwater-associated sills and dikes" by Zientek et al. (1985), occur in a 60-m-wide zone in metamorphic rocks below the base of the Stillwater Complex (Fig. 1) and have been studied in detail (Page, 1977, 1979; Zientek, 1983; Page et al., 1985; Helz, 1985). Geochemical studies show five major groupings based on variations in both major element and trace element compositions (Helz, 1985): (1) group 1, gabbronoritic diabase, (2) group 2, high-Mg gabbronorite, (3) group 3, mafic norite, (4) group 4, high-Ti noritic diabase, and (5) group 6, olivine gabbroic diabase. A group of noritic rocks (group 5) consists of samples of sills and dikes hybridized or contaminated by interaction with the enclosing metamorphic rocks. Petrographic and chemical data on samples from all groups revealed that recrystallization textures, which correlate with modal and chemical evidence for melt loss, were common in groups 2, 3, and 6; group 4 and the majority of group 1 samples were characterized by well-preserved igneous textures and bulk compositions (Helz, 1985, 1987). Although outcrops of the sills and dikes exist, none unambiguously illustrate their relation to the complex or to each other. Most of the field evidence indicates that the associated sills and dikes either predate or are contemporaneous with emplacement of the complex. Helz (1985) suggested that one or more of the magmas that fed the sills and dikes may represent a parent magma for the complex.

Our zircon study is intended (1) to date the various compositional groups of the sills and dikes to determine whether there is more than one age of intrusion and (2) to determine whether any of the sills and dikes are coeval with emplacement of the Stillwater Complex and may therefore be possible representatives of a Stillwater parent magma. We assume that emplacement and crystallization of the complex occurred in a short geologic time span. First-order approximations indicate maximum durations for cooling on the order of hundreds of thousands of years for a body approximately the size and shape of the Stillwater Complex. Such a short duration implies that zircon ages for the complex and for any sills and dikes derived from a Stillwater parent magma should be indistinguishable within the best possible error limits (~±1 Ma).

Note: Additional material for this article is Supplementary Data 9026, available on request from the GSA Documents Secretary (see footnote 1).

RESULTS

Brief descriptions of the sample minerals, location, and geochemistry as well as analytical methods, data, and age diagrams are in the GSA Data Repository.[1]

Associated Sills and Dikes

Seven zircon fractions were analyzed on a sample of group 1 gabbronorite; six were less than 1.5% discordant and one was about 17% discordant, yielding a concordia upper-intercept age of 2711 ±1 Ma. The more discordant analysis was composed of 13 whole, pinkish-purple to brown grains of a fairly magnetic zircon fraction; the other analyses represent the results from fractions containing the least magnetic zircons. The concordia lower-intercept age of 229 ±40 Ma for this sample is similar to other lower-intercept ages for zircons analyzed from samples of the Wyoming province (Premo and Van Schmus, 1989) and compares closely to the 211 ±134 Ma from a sample of group 1 diabase (Nunes, 1981).

Our sample of group 2 high-Mg gabbronorite is characterized by recrystallization textures that are generally indicative of melt loss. The absence of zircon in this sample may be due to melt loss, because Zr, an incompatible element, would have been concentrated in the expelled interstitial melt. In the absence of zircon and other accessory phases commonly used for U-Pb age determinations, a Pb-Pb isochron was constructed by using four separates (whole-rock, plagioclase, orthopyroxene, and clinopyroxene) and yielded an age of 2704 ±25 Ma. Sm-Nd results for the same separates yielded an isochron age of 2731 ±92 Ma and an initial $^{143}Nd/^{144}Nd$ ratio of 0.509054 ±115 ($\epsilon_{Nd} \approx -1.5 \pm 1.3$).

Nine analyses of zircons separated from a sample of group 3 mafic norite yielded a concordia upper-intercept age of 2703 ±10 Ma and a lower-intercept age of 190 ±100 Ma. Unlike the results from group 1, these analyses are significantly discordant (7% to 32%) and represent zircon fractions of very dark amber to brownish-purple, euhedral to subhedral grains. The increased discordance in this sample is a consequence of the higher U contents (3 to 4 times that in group 1) and, hence, higher degree of radiation damage, resulting in greater Pb loss at ca. 200 Ma. The zircon upper-intercept age is comparable to Sm-Nd internal isochron ages of 2689 ±40 Ma (Premo, unpublished data) and 2661 ±35 Ma (Snyder et al., 1987) for group 3 mafic norite. Although most group 3 samples are characterized by recrystallization textures, this particular sample of group 3 mafic norite is a good sample of quenched liquid and so contains zircon.

A regression of nine zircon analyses from a sample of group 4 high-Ti norite, which exhibits a well-preserved igneous texture, yielded a concordia upper-intercept age of 2712 ±3 Ma and a lower-intercept age of 240 ±210 Ma. The zircons from this sample are much smaller (less than 50 μm in length) than those from the previous samples and are typically subhedral to anhedral, light purple to clear, and nearly devoid of any inclusions, fractures, and discolorations.

As with group 2, our group 6 olivine gabbro sample showed recrystallization textures and other indications of melt loss; zircons as well as other accessory minerals like apatite were not found. Instead, isotopic analysis of separates (whole-rock, plagioclase, orthopyroxene, and olivine-pyroxene) yielded a Sm-Nd isochron age of 2706 ±64 Ma and an initial $^{143}Nd/^{144}Nd$ ratio of 0.509079 ±96 ($\epsilon_{Nd} = -0.74 \pm 1.1$).

Stillwater Complex

U-Pb data from zircon-baddeleyite for the Stillwater Complex are shown on a concordia diagram (Fig. 2). Zircons from a pegmatoidal anorthosite from the J-M Reef (JMR 3030′) of the Lower Banded series (TAZ 1) yielded two, small, hand-picked fractions. One fraction, 1(26), was composed of 21 pinkish-purple whole grains or fragments. Some of this material may have been baddeleyite, which was identified during scanning electron microscopy on similar samples from the J-M Reef. The presence of baddeleyite with zircon in these samples should not affect the U-Pb systematics. This fraction yielded a $^{207}Pb/^{206}Pb$ age of 2701 ±1 Ma at less than 3% discordance. The other fraction, 1(30), was composed of 21 whole

[1]These items, Supplementary Data 9026, are available on request from Documents Secretary, GSA, P.O. Box 9140, Boulder, CO 80301.

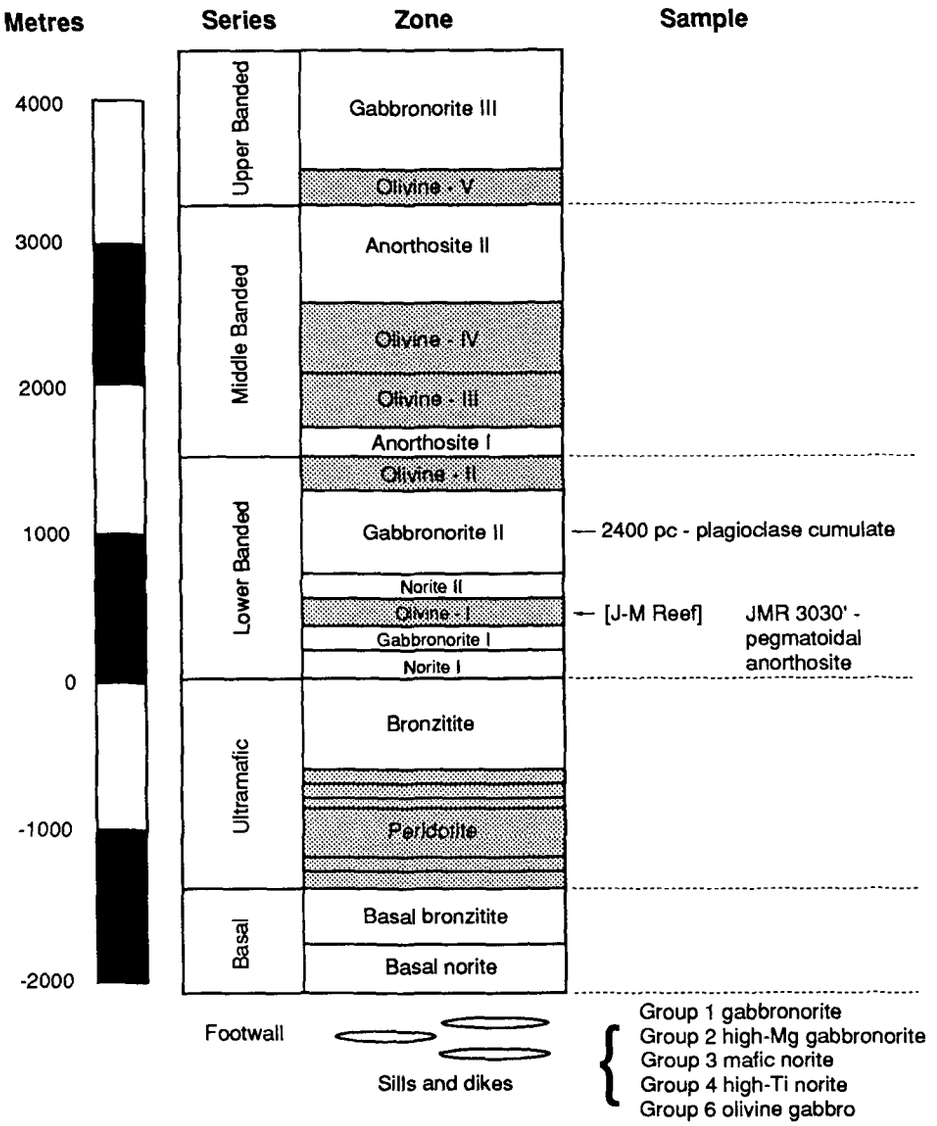


Figure 1. Generalized stratigraphic column of Stillwater Complex and associated sills and dikes that intrude footwall rocks (after Zientek et al., 1985; Page and Zientek, 1985). Stratigraphic height is measured relative to Ultramafic series–Lower Banded series contact. Approximate locations of samples analyzed in this study are shown on right side of column.

grains or fragments. A chord connecting the two analyses yields a concordia upper-intercept age of 2704 ±2 Ma and a lower-intercept age of 343 ±61 Ma.

Another sample from the Lower Banded series, 2400 pc, is a phlogopite-rich, mafic pegmatoid from a thin anorthositic layer near the base of gabbronorite zone II. Five fractions yielded a U-Pb zircon-baddeleyite, concordia upper-intercept age of 2707 ±5 Ma and a lower-intercept age of 630 ±300 Ma. The fractions were composed of elongate, large (~150 μm long), clear to purplish-brown, translucent to semiopaque, subhedral to euhedral zircon-baddeleyite grains. The analyses are slightly scattered and do not form a good linear array. Although we cannot justify picking and choosing among these analyses, we note that a chord drawn through three of the five fractions yields an upper-intercept age of 2705 ±1 Ma (mean standard weighted deviation [MSWD] = 0.011) that is nearly identical to the age defined by the two fractions from sample JMR 3030'. We believe that the striking linearity (shown by the very low MSWD) of these three analyses implies subtle heterogeneities in the two slightly deviant analyses, despite the near-concordance of one of them. Moreover, the lower-intercept age of 423 Ma for this regression is more agreeable with lower-intercept ages from the other samples (~200 ±200 Ma) compared to the 630 Ma result using all five analyses (assuming that the Pb loss in all samples was caused by the same disturbance).

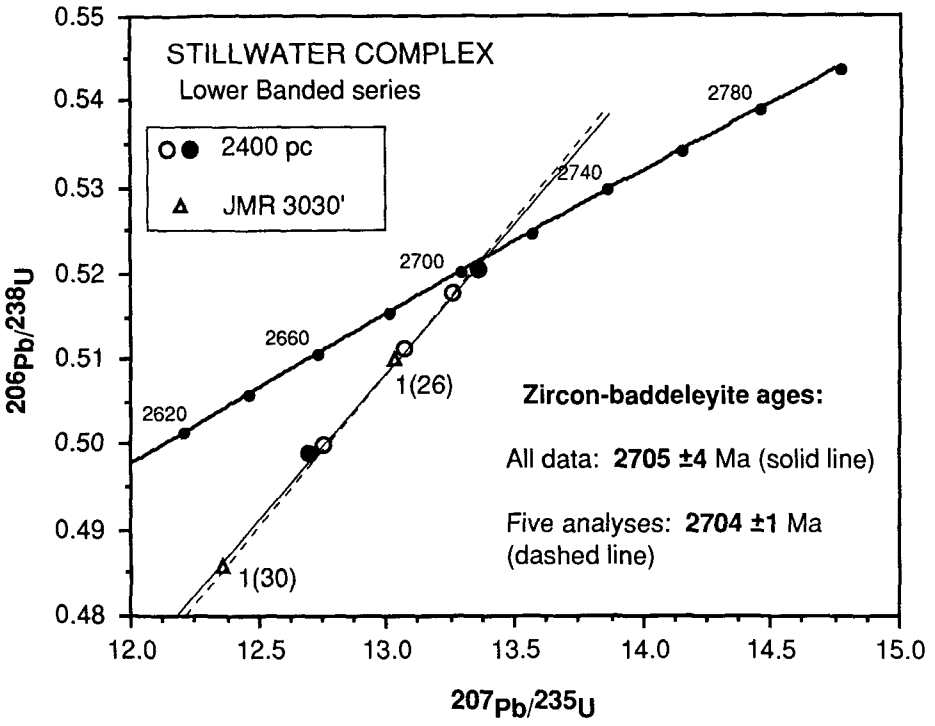


Figure 2. Concordia diagram of zircon-baddeleyite data (uncertainties at the 95% confidence level are smaller than size of symbols shown) from samples JMR 3030' (open triangles) and 2400 pc (open and solid circles) of Lower Banded series of Stillwater Complex. Combined results from seven analyses yield upper-intercept age of 2705 ±4 Ma. Omission of two analyses from 2400 pc (solid circles) alters age to 2704 ±1 Ma (43% probability of fit), determined by using regression treatment of Ludwig (1985b). Either age is in agreement with Sm-Nd age of 2701 ±8 Ma from DePaolo and Wasserburg (1979). Pb-isotope data for all samples were corrected with program of Ludwig (1985a) for mass fractionation, blank Pb (variable amounts between 40 and 100 pg of total Pb; blank Pb composition: $^{206}Pb/^{204}Pb$ = 18.8 ±0.6; $^{207}Pb/^{204}Pb$ = 15.65 ±0.2; and $^{208}Pb/^{204}Pb$ = 38.65 ±0.75), and initial common Pb by using values from Stacey and Kramers (1975) for age of sample.

COMBINED RESULTS

If we combine the results from the two Lower Banded series samples, we obtain a better linear fit of the data with a concordia upper-intercept age of 2705 ±4 Ma (Fig. 2). If we exclude the two slightly deviant analyses from 2400 pc, the other five fractions yield an upper-intercept age of 2704 ±1 Ma (MSWD = 0.93) and lower-intercept age of ~351 Ma. As argued above, the low MSWD for the linear fit of five out of seven analyses and the more agreeable lower-intercept age suggest to us that 2704 ±1 Ma is very close to the true age of the complex. In either case, both upper-intercept ages are in very good agreement with the Sm-Nd mineral isochron age of 2701 ±8 Ma (DePaolo and Wasserburg, 1979).

DISCUSSION

A comparison of the U-Pb zircon, zircon-baddelelyite, and Sm-Nd isochron ages for the Stillwater Complex and related rock units is illustrated in Figure 3. The zircon-baddelelyite age of 2705 ±4 Ma for the Lower Banded series of the Stillwater Complex is well within error and statistically indistinguishable from the Sm-Nd mineral isochron age of 2701 ±8 Ma (DePaolo and Wasserburg, 1979).

Zircon ages for groups 1 and 4 (2711 ±1 and

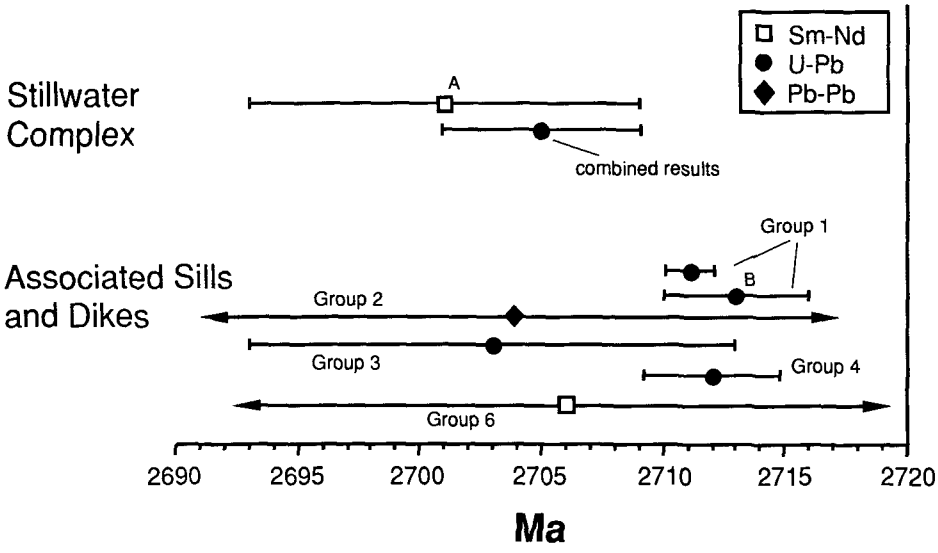


Figure 3. Diagram comparing precise radiometric age determinations for Stillwater Complex and related rock units. Ages for units are marked by symbols. Data of A from DePaolo and Wasserburg (1979); B from Nunes (1981). Error bars with arrows are open ended; errors for these ages are larger than scale shown. Whereas ages for groups 2, 3, and 6 are comparable to age for Stillwater Complex, other three ages for groups 1 and 4 do not overlap with Stillwater age within error limits, indicating that these two groups predate complex by 1 to 10 m.y.

2712 ±3 Ma, respectively) are in very good agreement with a U-Pb zircon age of 2713 ±3 Ma (Nunes, 1981) determined for a diabase sill (probably group 1 gabbronorite) and indicate that these groups are between 1 and 10 m.y. older than the Stillwater Complex (Fig. 3). These results are consistent with their well-preserved igneous textures, interpreted by Helz (1987) as indicating an older age.

If it is assumed that one or more of the sill and dike groups represent a parent magma (Helz, 1985), the most likely choices are groups 2, 3, and 6. Unfortunately, most available samples of these three chemical types have recrystallized and expelled a residual liquid during crystallization, as described in Helz (1987). Rocks affected this way (e.g., samples of groups 2 and 6 reported here) contain no primary zircon, because the expelled liquid contained most of the Zr present in the original magma. Although the ages for these groups are comparable to the age of the Stillwater Complex, the large errors (±25, ±10, and ±64 Ma for groups 2, 3, and 6, respectively) make it impossible to prove that these groups are coeval with the Stillwater Complex and do not negate the possibility that these groups are coeval with groups 1 and 4. Nevertheless, because this compaction melt-expulsion process would be favored by slow cooling (e.g., see McKenzie, 1985), such as must have occurred in the thermal aureole of the Stillwater Complex, it is probable that sills exhibiting such recrystallization behavior are closer in age to the Stillwater Complex than those that do not exhibit this texture, as hypothesized by Helz (1987).

These observations are consistent with the conclusions of Helz (1985) based on an extensive geochemical study of all the groups. In support of groups 2 and 3 being analogues for Stillwater liquids, Helz (1985) noted that (1) both groups have Mg/(Mg + Fe) ratios high enough to have produced olivine and bronzite cumulates with the compositions observed in the lower part of the Stillwater Complex; (2) pyroxenes from these groups have 10%–15% bulk normative Wo and are predominantly orthopyroxene, as in the lower part of the complex; (3) the bulk feldspar compositions of these rocks seem appropriate to produce plagioclase of An_{90-70} as a cumulus phase; (4) the rare earth element (REE) patterns commonly show steep light REE enrichments similar to those Lambert and Simmons (1987, 1988) inferred to be characteristic of a Stillwater liquid; and (5) small pods of mafic material with compositions very similar to groups 2 and 3 are found within cumulates of the Basal series (Zientek, 1983).

Group 6 olivine gabbro is also a candidate for a Stillwater parent magma. Although the mineralogic and chemical characteristics of group 6 are not consistent with production of the noritic and ultramafic lower sections of the complex, this group could represent the type of magma responsible for the olivine-rich zones (Fig. 1) that characterize the Banded series (Helz, 1985).

Group 1 gabbronorite has geochemical and isotopic characteristics suggesting it could be an analogue to a Stillwater magma; however, the zircon age of 2711 ±1 Ma clearly shows it is older than the Stillwater Complex. These rocks do show that magmatic activity related to the intrusion of the Stillwater Complex may have begun at least 2 m.y. (possibly 5 m.y., if the age of 2704 ±1 Ma given by five zircon-baddeleyite analyses from the Lower Banded series is accurate) before emplacement and crystallization of the complex.

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